Bargaining

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Game Theory and Mechanism Design - Lecture 10

Learning objectives

At the end of this lecture (and related tutorial(s)), a student should be able to:

- Understand why agents would **bargain**
- Explain the **rationale** behind bargaining (Nash programme)
- Formulate and discuss **Nash bargaining theorem**
- **Compute** Nash bargaining outcomes

Outline

1. Let's agree to disagree
2. Nash Bargaining
3. Nash's Bargaining solution
4. Nash axioms
5. Alternative bargaining solutions

1. **Let's agree to disagree**

Splitting the cake... again

There is 1 piece of cake left, to share between the 2 kids (cake sharing game)... The parent will only give them their share if they agree about it!

How would you do that?

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Well... we need to think a bit more about what is really going on here:

- we have an infinite number of Nash equilibria.
- worst-case they get nothing
- now, what if we say kid 1 values twice as much the quantity of cake than kid 2?

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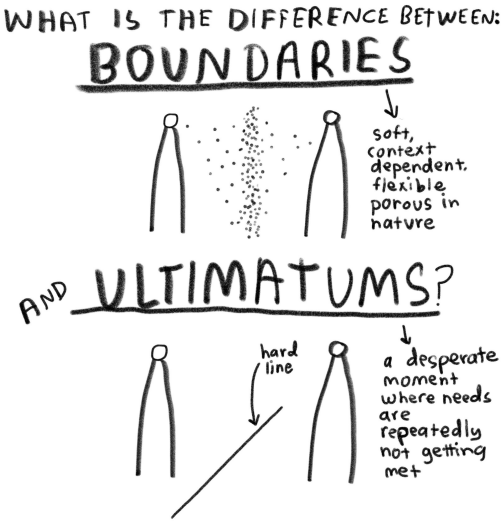
- now, what if we say kid 1 values twice as much the quantity of cake than kid 2?

Based on that, you get the idea!

The ultimatum game

Another basic game that often inspires the discussion about bargaining is the **ultimatum game**.



It is a general version of the cake sharing (or, inheritance sharing) game:

- Normal form game with 2 players ($N = \{1, 2\}$), as an equivalent to the original extended form game
- Their strategy is a value between 0 and 1, $s_i = [0, 1]$
- If $\sum_i s_i = 1$, their pay-off is s_i, and if $\sum_i s_i \neq 1$, the pay-off is 0.

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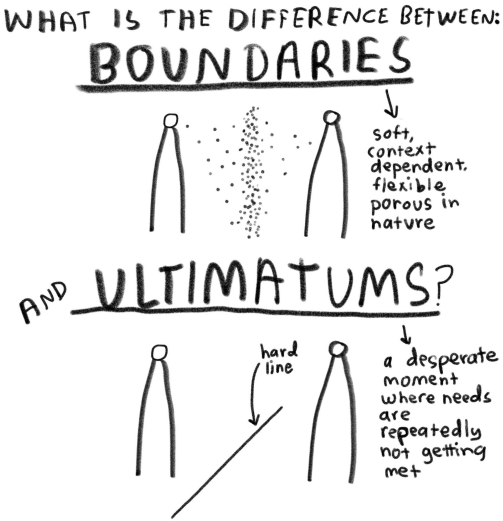

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- If $\sum_i s_i = 1$, their pay-off is s_i, and if $\sum_i s_i \neq 1$, the pay-off is 0.

By the way, what is their security value?

Why would they bargain?

Both players recognise that there must be a better *outcome* to that game!

Definition (Pareto dominance)

For a normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$ with strategy profiles $s_j \in \prod_i S_i$, a strategy profile s_k *Pareto-dominates* strategy profile s_j if the pay-offs from s_k are at least as good as those from s_j, for all players. I.e.

$$u_i(s_k) \geq u_i(s_j), \quad \forall i \in N\,,$$

with strict inequality for at least one of them.

In all games where some strategy profiles (and especially equilibria) are Pareto-dominated, there is room for bargaining...

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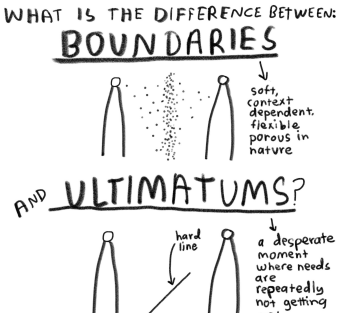


Do you see Pareto dominance and Pareto-dominated strategies in both cake-sharing and ultimatum games?

[*Note*: Nash equilibria necessarily Pareto-dominate strategy profiles that are not Nash equilibria...]

Nash program

Nash formulated this proposal even before to do his celebrated works on (Nash) equilibria(!)

Proposition (benefits from cooperation)

Any normal form game $\Gamma = \langle N, (S_i), (u_i) \rangle$ can be transformed into a more cooperative game, based on interaction, by extending the strategy sets of all players and allowing for correlated strategies.

Example from Lecture 9, based on the prisoner dilemma:

		player 2	
		NC	C
player 1	NC	$-4, -4$	$-6, 0$
	C	$0, -6$	$-5, -5$

original game

		player 2		
		NC	C	AC1
player 1	NC	$-4, -4$	$-6, 0$	$-6, 0$
	C	$0, -6$	$-5, -5$	$-5, -5$
	AC1	$0, -6$	$-5, -5$	$-4, -4$

game with extended strategy space

2. **Nash bargaining**

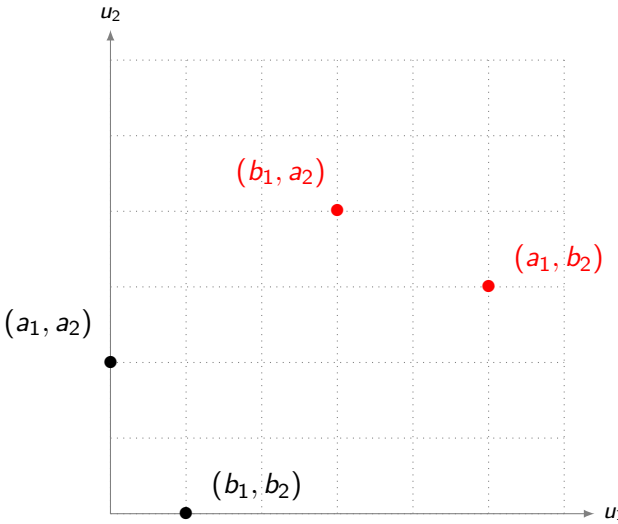
Visualising games

To understand Nash bargaining, we first need to *visualise* games...

player 2

player 1		a_2	b_2
	a_1	$0, 2$	$5, 3$
	b_1	$3, 4$	$1, 0$

(Nash equilibria are in red)

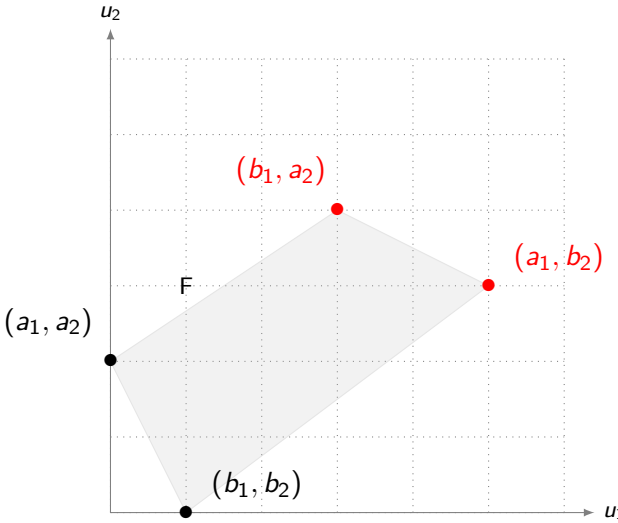


Feasibility set

Remembering contract and correlated strategies...

- a correlated strategy τ is a probability distribution over all potential combination of pure strategies (e.g., here, $\tau = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$)
- expected pay-offs are then a *convex* combination of the pay-offs for all strategy profiles
- if thinking of all potential probability distributions τ, the potential pay-offs are within a *convex hull* defined by the combination of strategies

This convex hull defines the *feasibility set* F for the bargaining process (since all these outcomes are possible to obtain, in principle)

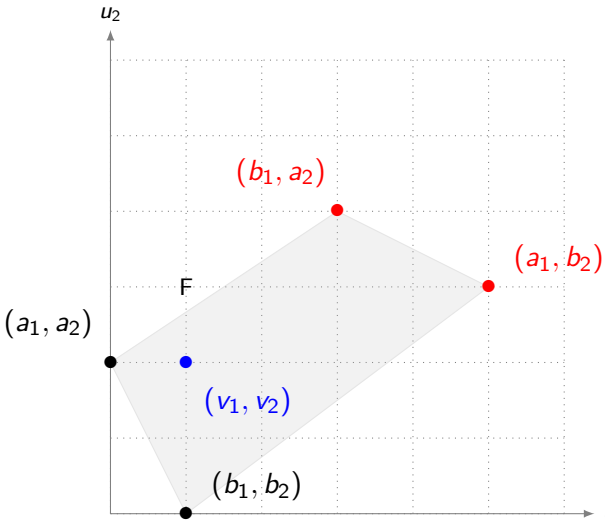


Disagreement point

We need to start from a necessarily Pareto-dominated outcome...

- the *disagreement point* (v_1, v_2) is a strategy profile that is necessarily Pareto-dominated, and for which we want to improve upon
- it also defines the minimum pay-offs players should get, whatever happens
- the combination of strategy profiles should always give us a feasible disagreement point!!

Security values defining the disagreement point yield $v = (v_1, v_2) = (1, 2)$

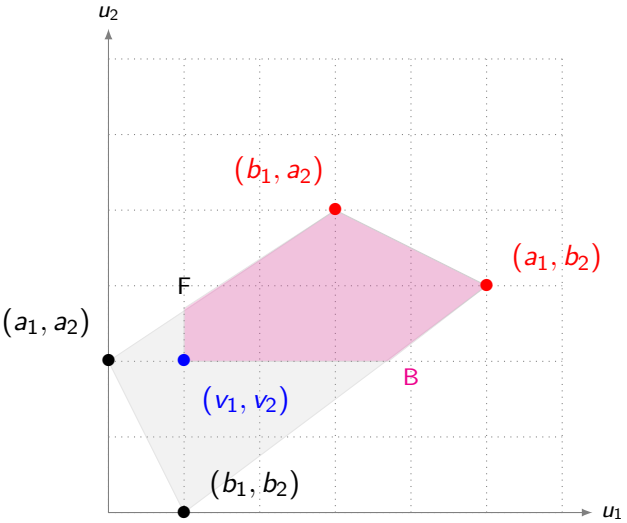


Reduced feasibility set

Individual rationality tells us that no outcome will be accepted if the pay-off of the players is below their security level...

- for player 1, $u_1(s_i) < v_1$, $\forall s_i \in \prod_i S_i$ is not acceptable
- for player 2, $u_2(s_i) < v_2$, $\forall s_i \in \prod_i S_i$ is not acceptable

This defines a reduced feasibility set
$B \subseteq F,\ B = \{(u_1, u_2) : u_1 \geq v_1,\ u_2 \geq v_2\}$.



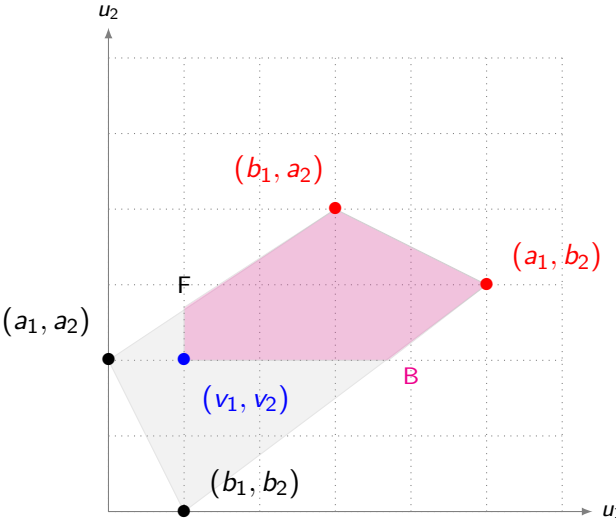
Which point to choose?

We know have a reduced feasibility set... All these points, in principle, can be an outcome of the bargaining process

Definition (2-person bargaining problem)

The *2-person bargaining problem* is defined by a feasibility set F and a disagreement point v. We consequently denote the bargaining problem (F, v).

- what outcome would you choose?
- what would make you choose that specific outcome?



3. **Nash's bargaining solution**

As an optimisation problem

Out of the (reduced) feasibility set, we have to pick a point that somewhat maximizes pay-offs for both players.

Definition (Nash's bargaining solution)

For a 2-person normal game, we can define an optimisation problem that gives a solution to the bargaining problem (F, v). That optimisation problem is

$$\begin{aligned}(u_1^*, u_2^*) \;=\; \underset{u_1, u_2}{\operatorname{argmax}} \quad & (u_1 - v_1)(u_2 - v_2) \\ \text{s.t.} \quad & (u_1, u_2) \in F \\ & u_1 \geq v_1, \;\; u_2 \geq v_2\end{aligned}$$

We also write $f(F, v) = (f_1(F, v), f_2(F, v))$ for that solution to the bargaining problem.

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We also write $f(F, v) = (f_1(F, v), f_2(F, v))$ for that solution to the bargaining problem.

Why do we (intuitively) multiply pay-offs instead of adding them?

Consider the case of the cake, with one agent that values twice as much the cake than the other...

- if we take the sum of pay-off in the optimisation problem, the best outcome is to give everything to the one who values the cake twice as much (and then nothing for the other one!)
- by multiplying, the joint pay-off is 0 if one of the player gets nothing...
- Eventually, the aim is to find the "sweet spot" between the best outcomes for both players involved.

Example application

Let us look again at out illustrative example.

- we want to maximise $(u_1 - 1)(u_2 - 2)$
- but we know that the solution should be on the line between (b_1, a_2) and (a_1, b_2), with equation

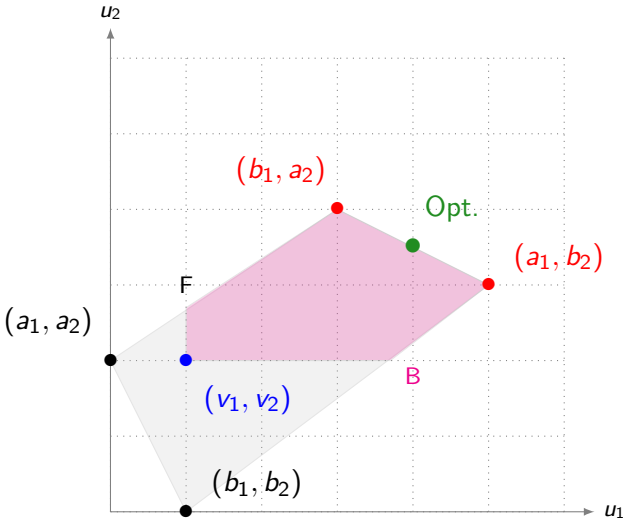
$$u_2 = -\frac{1}{2}u_1 + \frac{11}{2}$$

- we can plug that into the objective function, to make it a function of u_1 only

$$(u_1 - 1)\left(-\frac{1}{2}u_1 + \frac{7}{2}\right)$$

- To find the maximum, we differentiate and find the zero... this yields $u_1^* = 4$
- And, we can deduce that $u_2^* = \frac{7}{2}$.

Does that solution make sense to you? And, how would that be implemented?



4. **Nash axioms**

Nash's bargaining theorem

Nash's bargaining is (somewhat) the best one can do...!

Theorem (Nash's bargaining theorem)

Considering a 2-person bargaining problem (F, v), Nash's bargaining solution $(u_1^*, u_2^*) = f(F, v)$ is the only one with all the following properties:

- *strong Pareto efficiency*
- *individual rationality*
- *symmetry*
- *independence of irrelevant alternatives*
- *scale invariance (under positive linear transformations)*

These properties (also referred to as *axioms*) may sound a bit obscure... Let's go through them one by one.

[Note: individual rationality is obvious, since it is constraint in the optimisation problem (i.e., $u_i \geq v_i, \forall i$)]

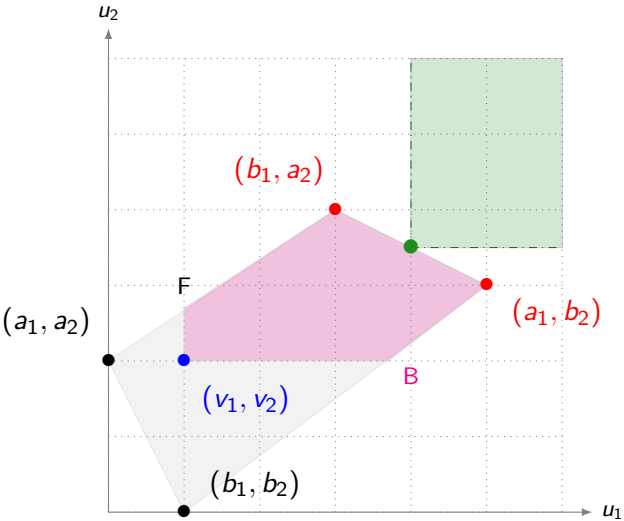
Strong Pareto efficiency

Definition (strong Pareto efficiency)

Given a feasible set F for the bargaining problem, a pay-off allocation $(u_1, u_2) \in F$ is said to be strongly Pareto efficient if there is no other allocation $(u_1', u_2') \in F$ such that $u_1' \geq u_1$ and $u_2' \geq u_2$, with strict inequality for at least one of the players.

Example in our case:

We cannot find a point in F for which any of the two players would get a strictly higher pay-off, with the other one not getting worse off.



Proposition (symmetry in Bargaining problem)

Given a feasible set F, that is symmetric for the 2 players, i.e.,

$$(u_1, u_2) \in F \implies (u_2, u_1) \in F$$

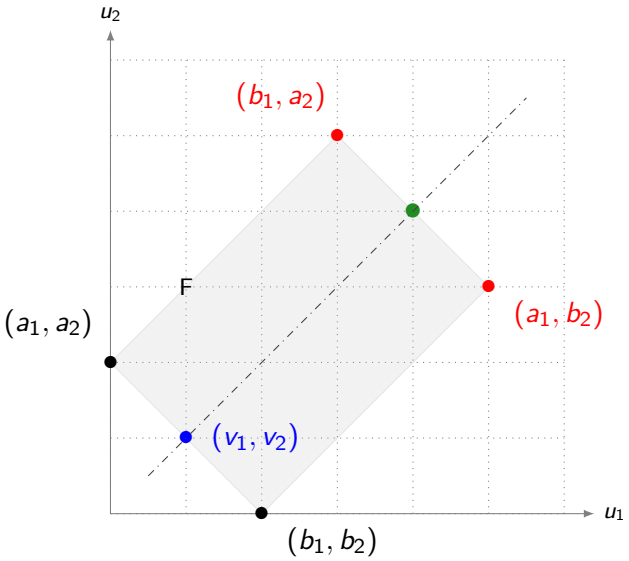
and symmetry in the disagreement point, i.e., $v_1 = v_2$, then the solution of Nash's bargaining problem is symmetric, $u_1^* = u_2^*$.

Example for another game:

We can see that

- the feasibility set is symmetric
- the disagreement point is symmetric
- Nash's bargaining solution is symmetric

[Note: for instance, the prisoner dilemma has this symmetry]



Independence of irrelevant alternatives

Proposition (independence of irrelevant alternatives)

Given a bargaining problem (F, v), if there is any other bargaining problem that is within (F, v), i.e., with

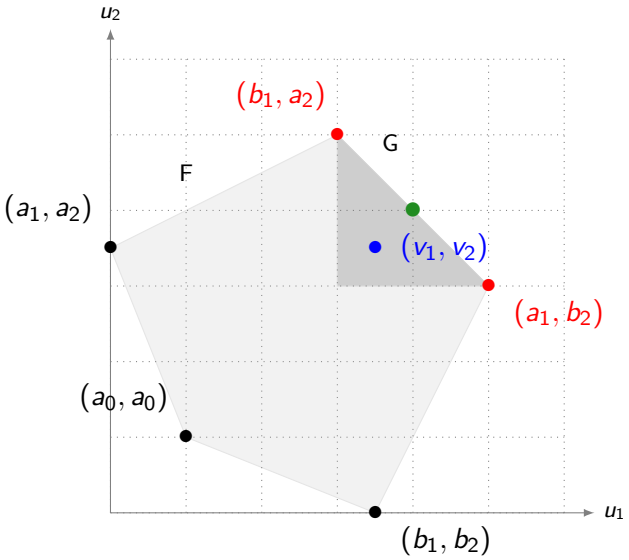
- $G \subseteq F$,
- $(v_1, v_2) \in G$ *and* $(v_1, v_2) \in F$,
- $f(F, v) \in G$

then these 2 bargaining problems have the same solution, $f(G, v) = f(F, v)$.

Example for another game:

We can see that

- $G \subseteq F$
- $(v_1, v_2) \in G$ and $(v_1, v_2) \in F$
- $f(F, v) = f(G, v)$



Scale invariance

Proposition (scale invariance)

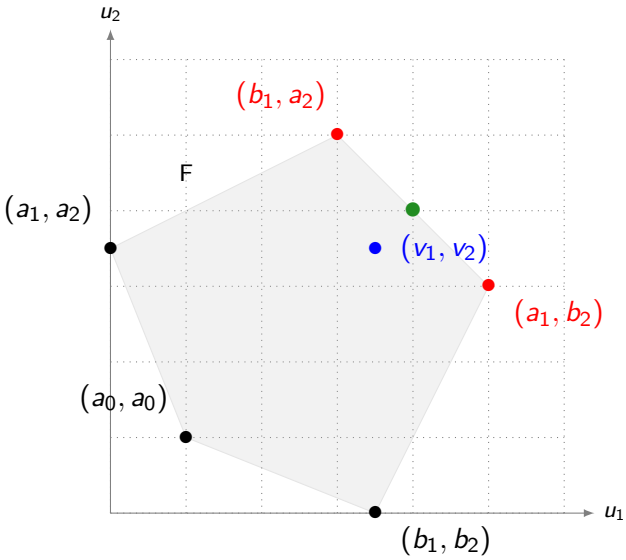
Given a bargaining problem (F, v), and if linearly transforming that problem to (G, w) such that

- $G = \{(\alpha_1 u_1 + \beta_1, \alpha_2 u_2 + \beta_2),\ (u_1, u_2) \in F\}$,
- $w = (w_1, w_2) = (\alpha_1 v_1 + \beta_1, \alpha_2 v_2 + \beta_2)$,
- $\alpha_1, \alpha_2, \beta_1, \beta_2 \geq 0$

then the solution of (G, w) follows the same linear transformation, i.e., $f(G, w) = (\alpha_1 u_1^* + \beta_1, \alpha_2 u_2^* + \beta_2)$ where $(u_1^*, u_2^*) = f(F, v)$.

Example with our other game:

Just look at the fact I do not use numbers for both axes... If you change the scale of axes (linearly), the solution is linearly scaled too...



5. **Alternative bargaining solutions**

Egalitarian and utilitarian solutions

The egalitarian solution concentrates on equal improvement of pay-offs.

Definition (egalitarian bargaining solution)

Given a bargaining problem (F, v), the *egalitarian solution* is given by

$$
\begin{aligned}
(u_1^*, u_2^*) \;=\; \underset{u_1, u_2}{\operatorname{argmax}} \quad & u_1 + u_2 \\
\text{s.t.} \quad & u_1 - v_1 = u_2 - v_2 \\
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And in the utilitarian case, we maximise pay-offs overall.

Definition (utilitarian bargaining solution)

Given a bargaining problem (F, v), the *utilitarian solution* is given by

$$\begin{aligned}(u_1^*, u_2^*) \;=\; \underset{u_1,u_2}{\operatorname{argmax}} \quad & u_1 + u_2 \\ \text{s.t.} \quad & (u_1, u_2) \in F, \quad u_1 \geq v_1,\; u_2 \geq v_2\end{aligned}$$

These 2 approaches to do meet all of Nash's axioms though (especially the scale invariance one)!

Wrapping up

- Bargaining is core to most **negotiation problems**
- We start from a **disagreement point**
- We build a **feasibility set** of solutions that can be implemented, e.g. through contracts
- The Nash's bargaining solution is the **unique solution** to the bargaining problem that has all important desirable properties

Further readings (/lecture notes for this topic):

Narahari Y (2014) *Game Theory and Mechanism Design* - chap. 26 (available on Blackboard Learn)

Thanks for your attention!